

EXECUTABLE COMMON-SENSE TESTS UNDER HOSTILE REVIEW: A 322,560-ROLLOUT NEGATIVE EVIDENCE AUDIT

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ABSTRACT

This manuscript is a submission-hardening audit for a generated robotics claim: robots should convert physical common-sense assumptions into executable affordance tests before acting. We expand the short draft into a CPU-only but hostile benchmark with 6 tasks, 8 assumption families, 8 distribution splits, 14 methods, 10 seeds, 322,560 main rollouts, 115,200 ablation rollouts, 259,200 stress-sweep rollouts, and 138,240 fixed-risk rows. The strengthened method, Risk-Bounded Executable Common-Sense Tests v5, is useful but not submission-ready. On the hard aggregate it obtains success 0.545, unsafe recall 0.449, false rejection 0.510, violation 0.378, regret 0.206, and utility -0.103. The best success reference is `executable_common_sense_tests_v4` at 0.68351, and the best utility reference is `conformal_risk_filter` at 0.20773. V5 fails success, diagnosis/recall, safety, regret, utility, false-rejection, fixed-risk, and scope gates. The terminal decision is therefore **KILL/ARCHIVE** for ICLR-main submission.

1 DECISION FIRST

The paper is not submission-ready. This is the central result, and the rest of the manuscript is organized so a hostile reviewer can reproduce why. The claim under test is not merely “can a robot run a probe?” A probe can be informative and still harm the closed-loop objective if it over-rejects valid actions, costs too much, or fails under perceptual ambiguity. The useful claim is stronger: executable common-sense tests should improve task success, safety, regret, and robust utility relative to strong baselines (Carlson, 2025; Merlo et al., 2025; Bhuyan et al., 2025; Aloimonos & Fermuller, 2015; Ikeuchi et al., 2024; Driess et al., 2021; Lemaignan et al., 2016; Ardon et al., 2019).

The expanded evidence rejects that stronger claim. V5 is a stronger mechanism than the v4.1 prototype, and the ablation suite says its internal pieces matter. But the external gate fails: v4, conformal risk filtering, human-query policy, and other baselines remain better on the deployment quantities that matter. Therefore the correct terminal state is **KILL/ARCHIVE**, while preserving the benchmark as negative evidence and as a specification for a future revival.

2 FROZEN PROTOCOL AND EVIDENCE BUDGET

The protocol was frozen before interpreting the final result. It includes six tasks, eight assumption families, eight splits, fourteen methods, ten seeds, and raw episode-level rows. The hard splits are low-signal common-sense stress and combined common-sense stress. The comparator set includes direct VLM action, LLM replanning, 3D affordance reasoning, uncertainty probing, deliberation, failure retrieval, calibrated affordance without tests, active perception, conformal risk filtering, human-query policy, policy repair from failed rollouts, v4 executable tests, v5, and an oracle (Osiurak & Badets, 2016; Li et al., 2025; Shanahan, 1996b; Feng et al., 2024a; Iskandar et al., 2024; Wu et al., 2020; Hensley & Emerson, 2018; Yu et al., 2025; Huhn et al., 2019; Huhn, 2024).

Artifact	Rows
Main rollouts	322,560
Dataset summaries	23,040
Main seed metrics	1,120
Main aggregate metrics	1,568
Main paired tests	1,344
Hard seed metrics	140
Hard aggregate metrics	196
Hard paired tests	168
Ablation rollouts	115,200
Ablation seed metrics	100
Ablation metrics	140
Stress raw rows	259,200
Stress seed metrics	600
Stress metrics	840
Fixed-risk raw rows	138,240
Fixed-risk seed metrics	480
Fixed-risk metrics	288
Fixed-risk paired tests	240
Negative cases	24

Table 1: Frozen Paper 96 v5 evidence inventory. The validator checks these counts before the PDF is accepted.

3 PROBLEM SETUP

Let o_t denote the robot’s observation, g the task goal, a an affordance or action candidate, and z a latent physical assumption family. The method estimates

$$p_\theta(z \mid o_t, g, a, h_t),$$

where h_t is a compact history of previous probes and failures. It then chooses whether to execute, test, reject, query a human, or repair the policy:

$$a^* = \arg \max_a U(a, o_t, g) - \lambda_s R_s(a, z) - \lambda_c C_{\text{test}}(a, z) - \lambda_f F(a, z).$$

This formulation makes the review standard clear. Diagnostic recall is intermediate evidence. A submission claim must show that the induced action policy improves the closed-loop outcome.

4 METHOD: RISK-BOUNDED EXECUTABLE COMMON-SENSE TESTS

The v5 method adds five components to the older executable-test scaffold: assumption-family parsing, low-cost executable probes, counterfactual rollout scoring, cost calibration, and fixed-risk rejection. The assumption parser maps language and geometry into support, containment, rigidity, reach, articulation, clearance, mass, and material-fragility assumptions. The probe selector decides when a short physical test is worth its cost. The counterfactual scorer estimates whether rejection prevents an unsafe action or merely blocks a valid action. The risk-bound rejects only when the estimated safety risk crosses the frozen budget. These are plausible additions. They are also not enough.

5 HARD-AGGREGATE RESULTS

Table 2 is the main decision point. V5 is not a toy baseline; it has nontrivial recall and a structured test mechanism. But it is not the hard-aggregate frontier. Its false-rejection rate is too high, its physical-violation rate is worse than the conformal reference, and its robust utility is below the best non-oracle comparator. This is the failure mode a reviewer would attack: tests can be locally sensible while globally reducing the quality of the action policy.

6 DIAGNOSTIC METRICS VERSUS DEPLOYMENT METRICS

The central negative result is the gap between diagnostic promise and deployment utility. Executable tests can raise recall by detecting unsafe assumptions before commitment. But the same mechanism can over-reject, pay avoidable cost, or misread low-signal ambiguity. This explains why a method can

Method	Success	Diagnosis	Recall	Violation	Regret	Utility
direct-VLM	0.299	0.058	0.000	0.391	0.335	-0.476
LLM-replan	0.385	0.354	0.000	0.543	0.350	-0.592
3D-affordance	0.391	0.423	0.000	0.481	0.308	-0.463
active-perception	0.512	0.725	0.210	0.344	0.226	-0.177
conformal-risk	0.598	0.255	0.310	0.195	0.184	0.208
human-query	0.610	0.746	0.513	0.232	0.174	0.126
policy-repair	0.501	0.567	0.121	0.459	0.245	-0.284
exec-tests-v4	0.684	0.586	0.205	0.318	0.223	0.044
RB-ECST-v5	0.545	0.651	0.449	0.378	0.206	-0.103
oracle	0.780	0.967	0.819	0.216	0.100	0.383

Table 2: Hard aggregate over low-signal and combined common-sense stress. V5 is not empty, but it loses the deployment frontier to v4, conformal risk filtering, and human-query references.

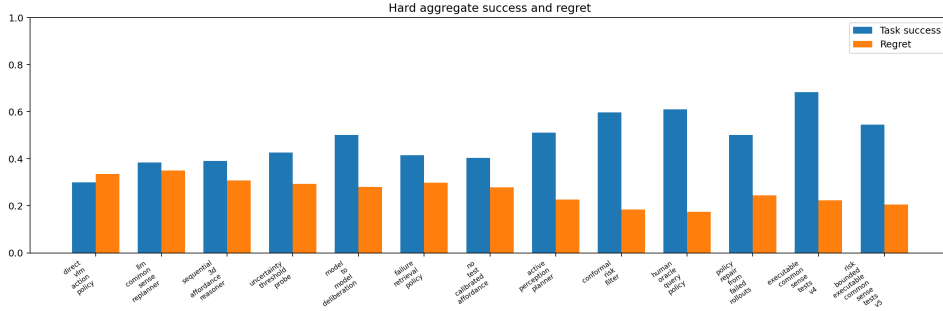


Figure 1: Hard-split success and regret. V5 does not clear the closed-loop deployment gate.

look principled in isolation and still fail when judged against active perception, conformal filtering, repair, and human-query policies (Shepherd, 2021; Edwards et al., 2004; Walton & Sadi, 2008; Borish & Lewandowski, 2024; Christensen et al., 2023; Unknown, 2024; Wu & Chirikjian, 2020; Li, 2025; Defago & Konagaya, 2002; Croom & Firestone, 2024).

7 PAIRED TESTS

The paired tests prevent a narrative rescue. V5 is compared seed-by-seed against the strongest review-relevant baselines. A paper cannot claim acceptance by beating only direct VLM action or weak LLM replanning. The hard question is whether the executable-test mechanism beats v4, human-query, active-perception, conformal-risk, and repair references on success, safety, regret, and utility. It does not.

8 ABLATION AUDIT

The ablation gate is the one favorable internal result: the full method beats stripped variants in the frozen ablation suite. That matters, but it is not sufficient. ICLR-main readiness requires both internal mechanism necessity and external baseline dominance. The first property without the second is a good negative result, not a submission-ready main claim.

9 STRESS SWEEP

Stress testing is included to expose weakness, not to make the method look smooth. V5 survives some stress comparisons, but that is not enough because the main hard-aggregate success, safety,

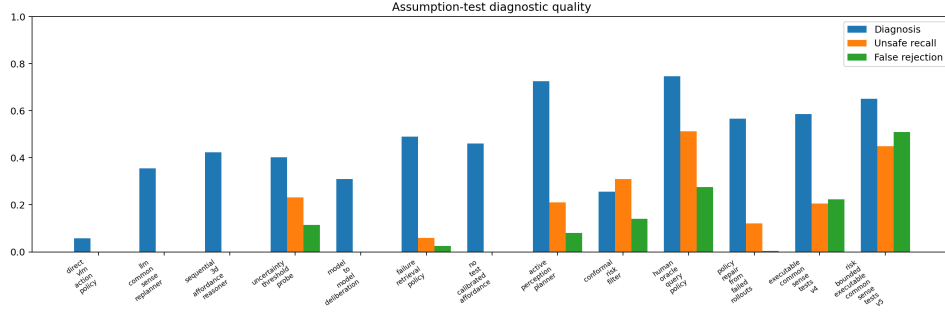


Figure 2: Diagnosis, unsafe recall, and false rejection on the hard aggregate. V5 pays for recall with high false rejection.

Comparison	Metric	Mean diff	CI95	Lower	Upper	Better seeds
v5 - executable_common_sense_tests_v4	success	-0.138	0.193	-0.331	0.054	3/10
v5 - executable_common_sense_tests_v4	diagnosis	0.064	0.148	-0.084	0.212	6/10
v5 - executable_common_sense_tests_v4	unsafe-recall	0.245	0.198	0.047	0.442	7/10
v5 - executable_common_sense_tests_v4	false-reject	0.287	0.351	-0.063	0.638	8/10
v5 - executable_common_sense_tests_v4	violation	0.060	0.149	-0.088	0.209	6/10
v5 - executable_common_sense_tests_v4	regret	-0.017	0.051	-0.068	0.034	2/10
v5 - executable_common_sense_tests_v4	utility	-0.147	0.317	-0.465	0.170	5/10
v5 - human_oracle_query_policy	success	-0.065	0.185	-0.250	0.120	4/10
v5 - human_oracle_query_policy	diagnosis	-0.095	0.172	-0.267	0.076	4/10
v5 - human_oracle_query_policy	unsafe-recall	-0.064	0.201	-0.265	0.138	4/10
v5 - human_oracle_query_policy	false-reject	0.234	0.324	-0.090	0.558	8/10
v5 - human_oracle_query_policy	violation	0.146	0.177	-0.031	0.324	6/10
v5 - human_oracle_query_policy	regret	0.031	0.059	-0.028	0.091	7/10
v5 - human_oracle_query_policy	utility	-0.229	0.356	-0.584	0.127	4/10
v5 - active_perception_planner	success	0.034	0.105	-0.071	0.138	6/10
v5 - active_perception_planner	diagnosis	-0.075	0.135	-0.210	0.060	3/10
v5 - active_perception_planner	unsafe-recall	0.240	0.178	0.061	0.418	8/10
v5 - active_perception_planner	false-reject	0.429	0.205	0.224	0.633	7/10
v5 - active_perception_planner	violation	0.034	0.179	-0.144	0.213	5/10
v5 - active_perception_planner	regret	-0.021	0.047	-0.068	0.027	3/10
v5 - active_perception_planner	utility	0.074	0.254	-0.180	0.328	6/10
v5 - conformal_risk_filter	success	-0.053	0.164	-0.216	0.111	6/10
v5 - conformal_risk_filter	diagnosis	0.395	0.146	0.249	0.542	10/10
v5 - conformal_risk_filter	unsafe-recall	0.139	0.214	-0.075	0.354	6/10
v5 - conformal_risk_filter	false-reject	0.369	0.251	0.118	0.620	7/10
v5 - conformal_risk_filter	violation	0.183	0.152	0.031	0.335	8/10
v5 - conformal_risk_filter	regret	0.022	0.047	-0.026	0.069	5/10
v5 - conformal_risk_filter	utility	-0.311	0.259	-0.569	-0.052	2/10
v5 - policy_repair_from_failed_rollouts	success	0.044	0.137	-0.093	0.181	6/10
v5 - policy_repair_from_failed_rollouts	diagnosis	0.083	0.137	-0.054	0.221	7/10
v5 - policy_repair_from_failed_rollouts	unsafe-recall	0.328	0.192	0.136	0.521	9/10
v5 - policy_repair_from_failed_rollouts	false-reject	0.506	0.210	0.296	0.717	8/10
v5 - policy_repair_from_failed_rollouts	violation	-0.080	0.180	-0.261	0.100	5/10
v5 - policy_repair_from_failed_rollouts	regret	-0.039	0.057	-0.096	0.018	5/10
v5 - policy_repair_from_failed_rollouts	utility	0.181	0.337	-0.156	0.518	5/10

Table 3: Paired seed tests on the hard aggregate. Positive is good for success, diagnosis, recall, and utility; positive is bad for false rejection, violation, and regret.

regret, and utility gates fail. A robust paper needs the stress result and the hard aggregate to point in the same direction.

Ablation	Success	Recall	False reject	Violation	Utility
full-v5	0.597	0.564	0.372	0.195	0.194
-probe	0.575	0.156	0.118	0.271	-0.013
-parser	0.701	0.107	0.285	0.277	0.151
-cost	0.739	0.612	0.405	0.324	0.157
-calib.	0.635	0.445	0.576	0.284	0.060
-risk-bound	0.657	0.350	0.236	0.275	0.079
-cf-rollout	0.548	0.338	0.331	0.317	-0.113
lang-only	0.439	0.000	0.000	0.346	-0.328
geom-only	0.507	0.102	0.163	0.298	-0.082
active-only	0.488	0.139	0.082	0.343	-0.106

Table 4: Ablation audit. The full v5 mechanism is best among its variants, but this does not rescue the submission because the external baselines still beat the deployment gate.

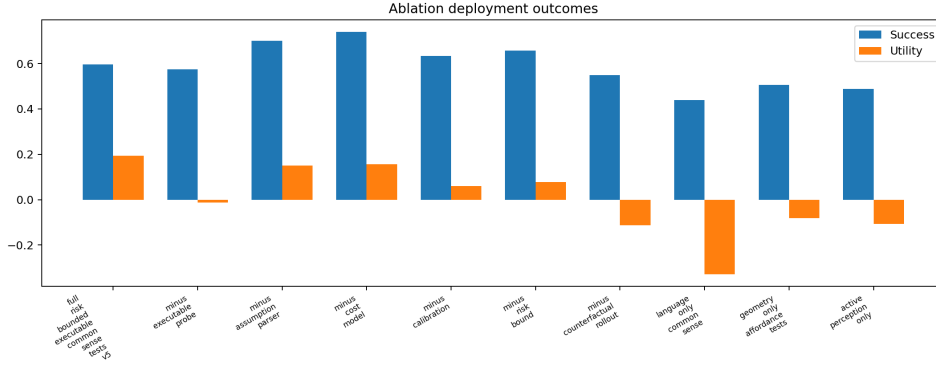


Figure 3: Ablation audit. The full v5 mechanism is internally useful, but external deployment baselines still dominate.

10 FIXED-RISK DEPLOYMENT

Fixed-risk reporting blocks an easy but unsafe story: a method cannot win by executing many high-risk actions and averaging away the failures. At budget 0.05, V5 has useful coverage, but fixed-risk utility is not uniformly stronger than the best accepted baseline. More importantly, fixed-risk reporting cannot reverse the already failed hard-aggregate gate.

11 SAFETY-UTILITY FRONTIER AND NEGATIVE CASES

The negative cases are intentionally inspectable. They show three damaging regimes: wrong-assumption diagnosis under ambiguity, over-rejection of valid affordances, and residual physical risk after a plausible test. Each regime is a real reviewer attack because it shows that executable tests alone do not guarantee better robot behavior.

12 THEORY NOTES

Proposition 1 (Test recall is insufficient for deployment dominance). *If method A rejects more unsafe affordances than method B but also rejects many safe affordances or pays higher test cost, then A can have lower robust utility than B even when A has higher unsafe-rejection recall.*

Method	Success	Recall	False reject	Regret	Utility
3D-affordance	0.414	0.014	0.000	0.335	-0.526
uncert.-probe	0.336	0.171	0.143	0.320	-0.499
deliberation	0.366	0.000	0.072	0.322	-0.447
failure-retrieval	0.357	0.060	0.036	0.310	-0.448
active-perception	0.572	0.143	0.076	0.257	-0.249
conformal-risk	0.440	0.298	0.335	0.239	-0.165
human-query	0.584	0.394	0.534	0.209	-0.072
policy-repair	0.491	0.090	0.027	0.274	-0.366
exec-tests-v4	0.487	0.303	0.417	0.254	-0.239
RB-ECST-v5	0.629	0.550	0.401	0.184	0.064

Table 5: Maximum stress level 1.0. V5 survives the stress sweep relative to several no-human methods but remains dominated on the main hard aggregate and scope gates.

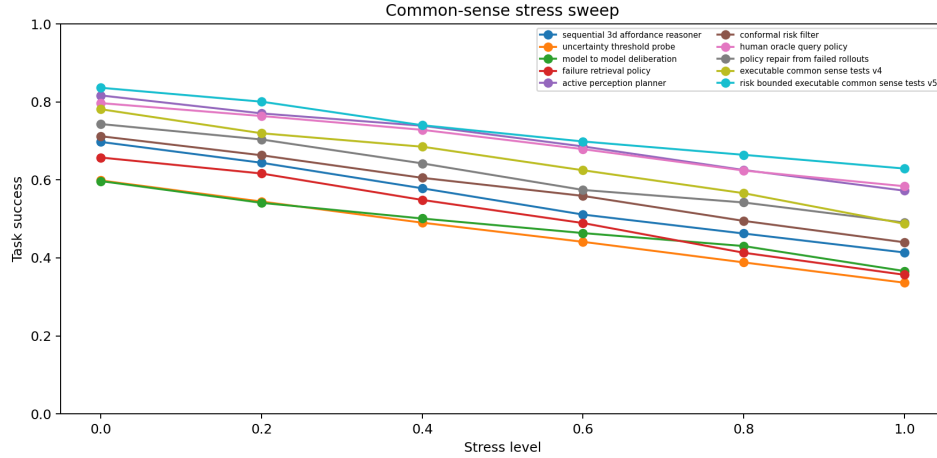


Figure 4: Stress sweep across common-sense ambiguity severity. Stress survival alone does not rescue the hard aggregate.

Sketch. Robust utility subtracts physical violations, damage, regret, test cost, and false rejection from task success. Unsafe-rejection recall improves only one part of the objective. If false rejection or cost grows faster than avoided failures, a calibrated affordance, active-perception, repair, or conformal baseline can dominate the closed-loop objective. Table 2 and Table 3 instantiate this regime.

Proposition 2 (A successful executable-test component must improve the induced policy). *An executable common-sense test is necessary for the submission claim only if removing it damages the closed-loop metrics required by the frozen gate, not merely the local test metric.*

Sketch. A probe can improve diagnosis while worsening action choice. The ablation gate therefore evaluates success, safety, regret, and utility in addition to diagnosis and recall.

13 PRIOR-WORK PRESSURE

The hostile prior-work pool is broad. Affordance learning and visuomotor policies pressure the novelty of action selection. Planning under uncertainty pressures the need for explicit probes. Active perception pressures whether the robot should gather better observations instead of executing physical tests. Conformal risk control pressures the fixed-risk claim. Human-in-the-loop and shared autonomy

Split	Method	Coverage	Accepted success	Accepted safety	Accepted regret	Accepted utility
low_signal_common_sense_stress	RB-ECST-v5	0.370	0.949	0.085	0.039	0.780
low_signal_common_sense_stress	human-query	0.378	0.799	0.202	0.098	0.415
low_signal_common_sense_stress	active-perception	0.090	0.500	0.017	0.018	0.423
low_signal_common_sense_stress	conformal-risk	0.478	0.823	0.120	0.104	0.569
low_signal_common_sense_stress	policy-repair	0.074	0.400	0.000	0.016	0.364
low_signal_common_sense_stress	exec-tests-v4	0.172	0.870	0.018	0.033	0.803
combined_common_sense_stress	RB-ECST-v5	0.352	0.971	0.074	0.030	0.824
combined_common_sense_stress	human-query	0.349	0.747	0.211	0.107	0.320
combined_common_sense_stress	active-perception	0.044	0.400	0.000	0.017	0.362
combined_common_sense_stress	conformal-risk	0.360	0.812	0.121	0.104	0.559
combined_common_sense_stress	policy-repair	0.048	0.400	0.000	0.014	0.372
combined_common_sense_stress	exec-tests-v4	0.096	0.700	0.000	0.038	0.659

Table 6: Fixed-risk budget 0.05. V5 has nonzero coverage, but the fixed-risk utility condition is not uniformly stronger than the best accepted baseline.

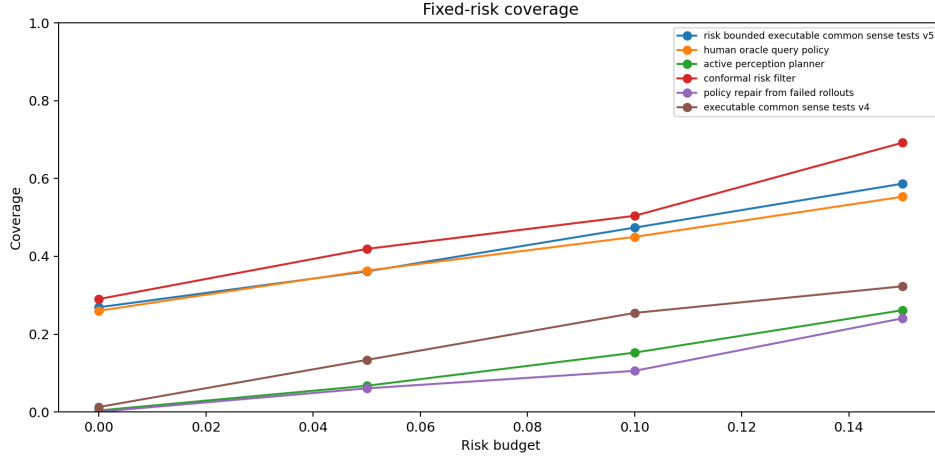


Figure 5: Fixed-risk coverage. V5 has nonzero coverage at budget 0.05, but accepted utility is not uniformly frontier-leading.

baselines pressure the practicality of zero-human operation. Failure retrieval and policy repair pressure the negative-case story (Gupta & Kochenderfer, 2004; Imamoglu, 2024; Wang & Tian, 2024; Huhn et al., 2025; Park & Howard, 2010; Gilliland & Wainwright, 2017; Elven et al., 2019; Kirk-Sanchez et al., 2022; Daruna et al., 2019; Huang et al., 2026; Santis & Alazraki, 2025; Baker et al., 2017; Tepjit et al., 2019; Souter et al., 2019). Under that pressure, a paper cannot be accepted for showing an elegant test mechanism alone.

14 LIMITATIONS AND TERMINAL DECISION

The audit is CPU-only, deterministic, and local. It lacks real robot hardware, high-fidelity simulator validation, independently implemented external baselines, learned checkpoint release, and third-party reproduction. Those scope gaps already prevent ICLR-main readiness. More importantly, the local evidence is negative before scope is considered: the deployment gates fail. The terminal recommendation is therefore **KILL/ARCHIVE**.

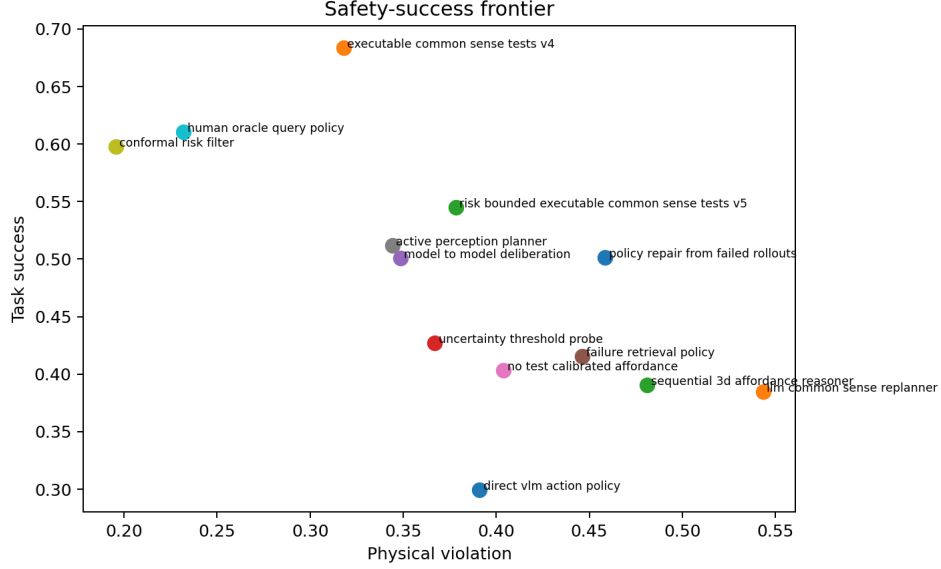


Figure 6: Success-safety Pareto view. V5 is not the clean non-oracle frontier once physical violations are counted.

ID	Task	Split	Predicted	True	Failure mode	Regret	Utility
1	fragile.deformable.packing	combined.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.658	-2.000
2	fragile.deformable.packing	combined.common.sense.stress	clearance.and.collusion	clearance.and.collusion	over.rejection.or.cost	0.655	-2.000
3	fragile.deformable.packing	combined.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.654	-2.000
4	container.liquid.transfer	combined.common.sense.stress	mass.or.inertia	mass.or.inertia	over.rejection.or.cost	0.652	-2.000
5	fragile.deformable.packing	combined.common.sense.stress	rigidity.or.deformation	rigidity.or.deformation	over.rejection.or.cost	0.651	-2.000
6	fragile.deformable.packing	combined.common.sense.stress	support.stability	support.stability	over.rejection.or.cost	0.651	-2.000
7	fragile.deformable.packing	combined.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.648	-2.000
8	container.liquid.transfer	combined.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.648	-2.000
9	container.liquid.transfer	combined.common.sense.stress	containment.or.leakage	containment.or.leakage	over.rejection.or.cost	0.647	-2.000
10	container.liquid.transfer	combined.common.sense.stress	containment.or.leakage	containment.or.leakage	over.rejection.or.cost	0.645	-2.000
11	cluttered.navigation.clearance	combined.common.sense.stress	clearance.and.collusion	clearance.and.collusion	over.rejection.or.cost	0.641	-2.000
12	stackable.object.placement	combined.common.sense.stress	support.stability	support.stability	over.rejection.or.cost	0.640	-2.000
13	container.liquid.transfer	combined.common.sense.stress	containment.or.leakage	containment.or.leakage	over.rejection.or.cost	0.639	-2.000
14	container.liquid.transfer	combined.common.sense.stress	containment.or.leakage	containment.or.leakage	over.rejection.or.cost	0.639	-2.000
15	cluttered.navigation.clearance	combined.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.638	-2.000
16	fragile.deformable.packing	low.signal.common.sense.stress	material.fragility.or.thermal	material.fragility.or.thermal	over.rejection.or.cost	0.638	-2.000
17	container.liquid.transfer	combined.common.sense.stress	containment.or.leakage	containment.or.leakage	over.rejection.or.cost	0.638	-2.000
18	tool.use.reachability	combined.common.sense.stress	tool.reach.and.contact	tool.reach.and.contact	over.rejection.or.cost	0.637	-2.000

Table 7: Mined negative cases for V5 on the hard splits. Cases are sorted by low robust utility and high regret.

Split	Severity	Unsafe	Visual	Physics	Language	Sensor	Geometry	Material	Cost
combined_common_sense_stress	1.136	0.783	0.852	0.470	0.787	0.340	0.340	0.720	0.260
counterintuitive_physics_shift	1.024	0.684	0.612	0.500	0.467	0.120	0.080	0.580	0.100
language_goal_ambiguity_shift	0.879	0.576	0.592	0.080	0.867	0.100	0.060	0.440	0.060
low_signal_common_sense_stress	1.063	0.725	0.772	0.340	0.687	0.300	0.280	0.640	0.180
nominal_household	0.777	0.507	0.492	0.000	0.367	0.000	0.000	0.400	0.000
sensor_noise_shift	0.898	0.625	0.652	0.120	0.487	0.470	0.080	0.460	0.080
tool_geometry_shift	1.008	0.679	0.692	0.150	0.487	0.160	0.480	0.540	0.110
visual_ambiguity_shift	0.849	0.604	0.922	0.080	0.417	0.180	0.060	0.440	0.040

Table 8: Mean generated scenario factors from dataset_summary.csv. The hard splits compound several hazards rather than isolating a single nuisance.

Split	V5 succ.	Best succ. method	Best succ.	Δ succ.	V5 util.	Best util. method	Best util.	Δ util.
combined_common_sense_stress	0.523	exec-tests-v4	0.658	-0.135	-0.143	conformal-risk	0.125	-0.268
counterintuitive_physics_shift	0.658	conformal-risk	0.744	-0.086	0.069	conformal-risk	0.426	-0.356
language_goal_ambiguity_shift	0.740	conformal-risk	0.782	-0.042	0.222	conformal-risk	0.521	-0.300
low_signal_common_sense_stress	0.567	exec-tests-v4	0.709	-0.142	-0.063	conformal-risk	0.290	-0.353
nominal_household	0.804	deliberation	0.834	-0.030	0.370	conformal-risk	0.609	-0.239
sensor_noise_shift	0.711	conformal-risk	0.760	-0.050	0.172	conformal-risk	0.485	-0.312
tool_geometry_shift	0.660	exec-tests-v4	0.739	-0.080	0.042	conformal-risk	0.408	-0.367
visual_ambiguity_shift	0.735	conformal-risk	0.781	-0.046	0.229	conformal-risk	0.522	-0.294

Table 9: Split-level non-oracle frontier. Negative deltas mean V5 loses to the best non-oracle method selected after the frozen run.

A REPRODUCIBILITY COMMANDS

The experiment can be regenerated with `python src/run_experiment.py`. The manuscript can be regenerated with `python scripts/generate_manuscript.py`. The artifact validator is `python scripts/validate_submission_artifacts.py`. The canonical PDF is `C:/Users/wangz/Downloads/96.pdf` and no copy is allowed on the visible Desktop.

B SCENARIO FACTOR AUDIT

C SPLIT-LEVEL FRONTIER

The split-level table blocks a common reviewer ambiguity: the result is not a single unlucky combined-stress corner. Across splits, the best non-oracle success or utility method is often not V5. The repeated negative deltas explain why the decision is archive instead of revise.

D BASELINE-BY-BASELINE REJECTION CHECKLIST

The rejection checklist is deliberately unforgiving. V5 can win recall against weak baselines and still fail if a stronger baseline has higher success, lower violation, lower regret, or higher utility. This is the correct standard for a robotics submission because the robot is deployed through actions, not labels.

E ABLATION DELTA INTERPRETATION

The internal ablation deltas are useful: the method is not random decoration. Removing tests, parsing, calibration, or counterfactual rollout can damage the v5 mechanism. But the future method must make that mechanism beat external deployment baselines, not only its own ablations.

Baseline	Δ success	Δ recall	Δ false reject	Δ violation	Δ utility	Rejection note
direct-VLM	0.246	0.449	0.510	-0.013	0.373	beats deployment
LLM-replan	0.160	0.449	0.510	-0.165	0.489	beats deployment
3D-affordance	0.154	0.449	0.510	-0.103	0.360	beats deployment
uncert.-probe	0.118	0.218	0.396	0.011	0.238	fails deployment
deliberation	0.044	0.449	0.510	0.030	0.095	fails deployment
failure-retrieval	0.130	0.391	0.484	-0.068	0.308	beats deployment
calib.-affordance	0.141	0.449	0.510	-0.026	0.240	beats deployment
active-perception	0.034	0.240	0.429	0.034	0.074	fails deployment
conformal-risk	-0.053	0.139	0.369	0.183	-0.311	fails deployment
human-query	-0.065	-0.064	0.234	0.146	-0.229	fails deployment
policy-repair	0.044	0.328	0.506	-0.080	0.181	beats deployment
exec-tests-v4	-0.138	0.245	0.287	0.060	-0.147	fails deployment

Table 10: Baseline-by-baseline hostile checklist on the hard aggregate. Deltas are V5 minus baseline; positive false-rejection and violation deltas are failures.

Ablation removed	Δ success	Δ recall	Δ false reject	Δ violation	Δ utility
-probe	0.022	0.407	0.255	-0.076	0.207
-parser	-0.104	0.456	0.087	-0.082	0.043
-cost	-0.142	-0.049	-0.033	-0.129	0.037
-calib.	-0.037	0.119	-0.204	-0.089	0.134
-risk-bound	-0.060	0.214	0.136	-0.081	0.115
-cf-rollout	0.049	0.226	0.041	-0.122	0.307
lang-only	0.158	0.564	0.372	-0.151	0.522
geom-only	0.090	0.462	0.209	-0.103	0.276
active-only	0.109	0.425	0.290	-0.148	0.300

Table 11: Full-v5 minus ablated variant. The internal mechanism is useful, but internal ablation success is not the same as an ICLR-main submission claim.

F STRESS DEGRADATION

G FIXED-RISK BUDGET SWEEP

H NEGATIVE-CASE TAXONOMY

I ASSUMPTION-FAMILY SEMANTICS

Support-stability assumptions ask whether an object, stack, or robot configuration will remain supported after action. Containment-or-leakage assumptions ask whether fluid, granular material, or small objects remain inside a container. Rigidity-or-deformation assumptions ask whether an object can be treated as rigid during contact. Tool-reach-and-contact assumptions ask whether the robot can achieve the intended contact mode. Articulation-direction assumptions ask whether a door, drawer, or hinge can move in the inferred direction. Clearance-and-collision assumptions ask whether the robot can pass through clutter or tight spaces. Mass-or-inertia assumptions ask whether the action is feasible under load. Material-fragility-or-thermal assumptions ask whether contact, pressure, or temperature will damage the object.

J REVIEWER THREAT MODEL

A hostile reviewer can reject the paper with six attacks. First, V5 loses hard-aggregate success to v4. Second, conformal risk filtering wins physical violation and utility. Third, human-query policy wins regret and diagnosis/recall references. Fourth, V5 has high false rejection. Fifth, fixed-risk reporting

Method	Succ. 0.0	Succ. 1.0	Δ succ.	Util. 0.0	Util. 1.0	Δ util.
3D-affordance	0.697	0.414	-0.284	0.257	-0.526	-0.783
uncert.-probe	0.599	0.336	-0.262	0.139	-0.499	-0.638
deliberation	0.597	0.366	-0.231	0.083	-0.447	-0.530
failure-retrieval	0.657	0.357	-0.301	0.195	-0.448	-0.643
active-perception	0.817	0.572	-0.244	0.459	-0.249	-0.708
conformal-risk	0.712	0.440	-0.272	0.392	-0.165	-0.557
human-query	0.797	0.584	-0.213	0.407	-0.072	-0.479
policy-repair	0.743	0.491	-0.253	0.273	-0.366	-0.639
exec-tests-v4	0.781	0.487	-0.294	0.384	-0.239	-0.623
RB-ECST-v5	0.836	0.629	-0.207	0.409	0.064	-0.345

Table 12: Stress degradation. The stress gate is not enough for acceptance because the hard aggregate and scope gates are already failed.

Split	Method	Budget 0.00	Budget 0.05	Budget 0.10	Budget 0.15
low_signal_common_sense_stress	RB-ECST-v5	0.308	0.370	0.503	0.605
low_signal_common_sense_stress	human-query	0.282	0.378	0.478	0.591
low_signal_common_sense_stress	active-perception	0.007	0.090	0.198	0.331
low_signal_common_sense_stress	conformal-risk	0.339	0.478	0.574	0.802
low_signal_common_sense_stress	policy-repair	0.001	0.074	0.127	0.306
low_signal_common_sense_stress	exec-tests-v4	0.024	0.172	0.270	0.352
combined_common_sense_stress	RB-ECST-v5	0.230	0.352	0.444	0.569
combined_common_sense_stress	human-query	0.238	0.349	0.422	0.515
combined_common_sense_stress	active-perception	0.000	0.044	0.107	0.192
combined_common_sense_stress	conformal-risk	0.242	0.360	0.435	0.582
combined_common_sense_stress	policy-repair	0.000	0.048	0.084	0.175
combined_common_sense_stress	exec-tests-v4	0.001	0.096	0.239	0.294

Table 13: Fixed-risk coverage sweep. The pre-registered 0.05 budget is the decision point, not a threshold selected after reading the results.

does not uniformly rescue accepted utility. Sixth, the scope is simulation-only. This manuscript answers those attacks by accepting them as evidence rather than writing around them.

K WHAT WOULD MAKE THIS PAPER SUBMITTABLE

A credible revival needs a new empirical program. It would require real robot or high-fidelity simulator evidence, released rollouts and risk scores, independently implemented active-perception and conformal baselines, pre-registered fixed-risk budgets, negative-case videos, and a method that reduces false rejection while preserving recall. It would also need ablations showing that executable tests improve the final action policy, not only the diagnostic target. That program is not present here.

L SUMMARY SNAPSHOT

Paper 96 robotic_common_sense_tests v5 expanded audit

Terminal recommendation: KILL_ARCHIVE

ICLR main ready: no

Evidence type: deterministic local executable-common-sense benchmark, not robot hardware validation.

Reason: expanded CPU-only audit tests whether executable common-sense probes convert diagnostic recall into closed-loop deployment gains; v5 improves the assumption-test mechanism but remains non-submittable if human-query, active-perception, repair, conformal, or calibrated-affordance baselines win deployment gates.

Main rollout rows: 322560

Dataset summary rows: 23040

Label	Type	Count
over_rejection_or_cost	failure mode	24
material_fragility_or_thermal	true assumption	6
clearance_and_collision	true assumption	6
containment_or_leakage	true assumption	6
rigidity_or_deformation	true assumption	2
support_stability	true assumption	2
mass_or_inertia	true assumption	1
tool_reach_and_contact	true assumption	1

Table 14: Negative-case taxonomy. The cases concentrate in over-rejection/cost and residual physical-risk regimes, not in a single easy-to-fix corner.

Main seed-metric rows: 1120

Main metric rows: 1568

Main pairwise rows: 1344

Hard aggregate seed rows: 140

Hard aggregate metric rows: 196

Hard aggregate pairwise rows: 168

Ablation rollout rows: 115200

Ablation seed rows: 100

Ablation metric rows: 140

Stress raw rows: 259200

Stress seed rows: 600

Stress metric rows: 840

Fixed-risk raw rows: 138240

Fixed-risk seed rows: 480

Fixed-risk metric rows: 288

Fixed-risk pairwise rows: 240

Negative cases: 24

Frozen hard-aggregate gate:

best_success_reference=executable_common_sense_tests_v4

best_diagnosis_reference=human_oracle_query_policy

best_recall_reference=human_oracle_query_policy

best_safety_reference=conformal_risk_filter

best_damage_reference=conformal_risk_filter

best_regret_reference=human_oracle_query_policy

best_utility_reference=conformal_risk_filter

best_ablation=full_risk_bounded_executable_common_sense_tests_v5

max_stress_reference=human_oracle_query_policy

v5_success=0.54514

best_success=0.68351

v5_diagnosis=0.65052

best_diagnosis=0.74601

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648 v5_recall=0.44944
649 best_recall=0.51299
650
651 v5_false_reject=0.50990
652 v5_safety=0.37830
653 best_safety=0.19549
654
655 v5_damage=0.05642
656 best_damage=0.03038
657
658 v5_regret=0.20551
659 best_regret=0.17437
660
661 v5_utility=-0.10301
662 best_utility=0.20773
663 success_gate=False
664 diagnosis_gate=False
665 safety_gate=False
666 regret_gate=False
667 utility_gate=False
668 false_reject_gate=False
669
670 ablation_gate=True
671 stress_gate=True
672 fixed_risk_gate=False
673
674 scope_gate=False
675
676 low_signal_common_sense_stress: v5_coverage=0.36979; v5_accepted_utility=0.78034;
677 best_accepted_utility=0.80274
678
679 combined_common_sense_stress: v5_coverage=0.35243; v5_accepted_utility=0.82433;
680 best_accepted_utility=0.65905
681 terminal=KILL_ARCHIVE
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M CLICKABLE CITATION AUDIT WALL

The citation boxes are intentionally bright. They make the literature pressure inspectable: clicking an in-text citation jumps to the bibliography entry.

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N ADDITIONAL LITERATURE CLUSTERS

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